Satya-Score: An AI Hallucination Detection & Scoring Framework

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***Abstract*—Large language models (LLMs) state false facts fluently, and in Indian languages these errors are rarely checked. We present Satya-Score, a claim-level fact-checking system for English and Hindi LLM output. It splits text into atomic claims and routes each claim to the most authoritative source that can decide it: official statistics from the World Bank and the WHO and structured facts in Wikidata first, then professional fact-checks, and otherwise Wikipedia passages judged by a multilingual natural-language-inference (NLI) model. Every verdict (Verified, Uncertain, Hallucinated) links its evidence and, when a structured source decided it, states the correct fact. Most design changes were tested against decision rules fixed before each run. Structured sources decide almost without error (11 of 13 knowledge-graph decisions correct in the pre-registered run and 12 of 12 after a linking fix; 137/137 false statistics caught with no true claim contradicted on 40 randomly chosen countries), and a negation-safe fact-check rule made 0 wrong decisions. On real chatbot claims judged from Wikipedia text alone the system rarely flags true claims (3.5% false alarms) but catches few hallucinations; an analysis shows why NLI reads topical overlap as support, and that an LLM judge catches more at nearly four times the false-alarm rate.**

***Index Terms*—hallucination detection, fact verification, natural language inference, knowledge graphs, Hindi, cross-lingual retrieval, large language models.**

I. Introduction

LLM chatbots are increasingly used for health, legal and government information, where an undetected false statement can cause harm. Errors are usually local: a ten-sentence answer may contain one wrong date or name. Checking must therefore happen claim by claim, and each verdict must be traceable to evidence a reader can open. Most existing checkers are English-only, although hundreds of millions of users read AI answers in Hindi and other Indian languages.

This work builds and evaluates a bilingual (English–Hindi) claim-level checker. Its contributions are:

1) A source-routing architecture that sends each claim to the most authoritative available source (fact-check, official statistics, knowledge graph, encyclopaedic text) and explains every verdict with its source and, when a structured source decides, the correct fact.
2) A negation-safe rule for applying professional fact-checks: a rating carries over only when the NLI model finds the required entailment direction and no contradiction.
3) Two-agency agreement for official statistics, comparing figures at the precision they are written to.
4) Hindi knowledge-graph verification with relation templates, exact-name linking and a namesake rule.
5) An evaluation with decision rules fixed in advance that reports negative results: stricter evidence rules, a number guard, a second evidence source, larger NLI models and an LLM judge were each tested, and the four decisions that went against a rule's outcome are reported.

II. Related Work

FEVER [1] established claim verification against Wikipedia. FActScore [2] and SAFE [3] decompose long-form LLM output into atomic facts and verify them, SAFE with an LLM agent and web search, agreeing with human annotators 72% of the time. Purpose-built grounding checkers such as AlignScore [4], MiniCheck [5] and FactCG [6] reach about 75% balanced accuracy on the LLM-AggreFact benchmark, close to GPT-4. Real-world claim verification remains hard: the best AVeriTeC 2024 system scored 63% using GPT-4 [7]. Multilingual resources include X-FACT [8] and the Mu-SHROOM task [9], whose strongest Hindi systems used very large LLMs. NLI models are known to treat lexical overlap as entailment [10]. Our system combines structured sources (Wikidata [11]) with cross-lingual retrieval (XLM-R [12], multilingual sentence embeddings [13], FAISS [14]).

III. System Design

A. Claim Extraction

English answers are parsed with spaCy; compound sentences are decomposed, list items become claims, and pronouns are replaced by the sentence topic rather than the nearest name (on 400 held-out Wikipedia introductions this raised correct referents from 93 to 277). Questions, pleasantries, advice and opinions are skipped. Hindi is split at the sentence level.

B. Evidence Retrieval

Claims and passages are embedded with a multilingual XLM-R sentence encoder [12], [13] and searched in a FAISS index [14] of 102,069 Wikipedia passages (89,594 English, 12,475 Hindi). For each claim the system also searches Wikipedia live and adds missing articles without rebuilding the index. The top ten passages above a relevance threshold are judged.

C. NLI Judgement and the Evidence Rule

A multilingual NLI model (mDeBERTa-v3-base-mnli-xnli) scores each passage. Support decides first; only a passage whose article is about the claim’s subject may contradict it, otherwise the claim is Uncertain. This rule cut false alarms on fresh true statements from 33 to 8 of 99, but it also lowered the hallucinations caught on training claims from 11 to 3 of 45 and failed one of its pre-fixed checks; it was adopted as a product decision.

D. Structured Verification

Claims of known types are decided by structured sources. Wikidata checks places, capitals, dates of birth, death and founding, authorship, current office-holders, heights, lengths, populations, areas and currencies, in English and Hindi; subjects are matched by exact name, and with namesakes any match supports while a contradiction needs every namesake with a value to disagree. Country statistics (population, GDP, life expectancy, literacy, infant mortality) are checked against World Bank and WHO data; where both agencies report a figure they must agree. Professional fact-checks are retrieved through the Google Fact Check Tools API and applied only under the negation-safe rule.

E. Scoring, Explanation and Implementation

Per-claim scores follow the composite formula HalluScore = $\alpha \cdot \text{SemSim} + \beta \cdot \text{NLI} + \gamma \cdot \text{KG}$. Similarity measures topic rather than truth, so α is fixed at 0, and β and γ (about 0.5 each) are trained on labelled claims. A structured-source decision sets the score (1.0 or 0.0); otherwise a matching fact-check sets it (0.05 or 0.95); otherwise the NLI evidence score is used. Scores below 0.35 are Hallucinated and above 0.65 Verified. The system is a FastAPI service with a React web application, accounts and history. Parallel, per-service-paced lookups reduced a six-claim check from 27.7 s to 5.1 s on first sight and to 0.7 s when repeated, on a laptop GPU, with identical verdicts.

IV. Evaluation Methodology

The labelled data are 456 claims from LLM answers (266 English, 190 Hindi) in five domains (government, health, science, sports, history), and 40 relational claims from two further LLMs. The full set was labelled by a team of four annotators (the authors), who shared the claims and reviewed AI-suggested verdicts; two annotators labelled a blind sample of 56 claims with binary agreement $\kappa = 0.53$ (raw agreement 88%). A grouped split keeps claims from the same question in one partition; binary evaluation uses 303 training and 76 test claims. Fresh true statements are lead sentences of unseen Wikipedia articles. Changes were tested against decision rules written and committed before the run, except two exploratory knowledge-graph analyses; product decisions that overrode a rule are recorded.

V. Results

TABLE I
LIVE SYSTEM ON THE ANNOTATED AND FRESH SETS

Set	Correct	False alarms	Caught
Train (303)	157	9 / 258	5 / 45
Test (76)	38	3 / 66	0 / 10
Fresh true sentences (128)	113	4 / 128	—

TABLE II
STRUCTURED SOURCES AND OTHER COMPONENTS

Component (test data)	Result
Wikidata decisions, pre-registered	11 / 13 correct
Same, after linking fix (exploratory)	12 / 12 correct
Statistics, 40 random countries	137 / 137 false caught; 0 true contradicted
Fact-checks (25 rumours, 15 true)	0 wrong; rumours caught 3 → 10
KG ablation, pre-registered (40)	AUC 0.704 → 0.784
Round 3, KG deciding (40)	correct 14 → 24 of 40
Pronouns (400 held-out texts)	right referents 93 → 277

When the system commits to a verdict on real chatbot claims, about four in five verdicts are correct; most errors are hallucinations marked Verified. An analysis of the 45 training hallucinations found that about 31 were artifacts of an earlier claim extractor rather than chatbot errors, and that most of the remaining genuine errors are misconceptions and over-generalisations that no structured source covers. Eight of these were Verified by passages that are only topically related, and on fifteen common-sense statements the system Verified all nine false ones; a purpose-built grounding checker (MiniCheck) did not remove this failure in a design probe.

TABLE III
CHANGES TESTED AND DECISIONS TAKEN

Change	Outcome and decision
Top-10 passages	correct 130→170/303; false alarms 8→11; adopted
Larger NLI models	rule favoured them; within noise; kept current
Number guard (2 versions)	lost true Verified claims
MedlinePlus as 2nd source	false alarms 9 → 11
Strict / capped evidence rules	true Verified 152 → 47 / 107
LLM judge (open mode)	caught 5→18 of 45; false alarms 9→35; kept off

VI. Discussion and Limitations

The results separate two regimes. Claims of a type a structured source records are decided reliably and explained with the correct fact. General statements judged from encyclopaedic text are the weak point: an NLI model cannot distinguish a passage that states a fact from one that merely discusses it, and refuting evidence for common-sense falsehoods is rarely written explicitly. The only method that caught substantially more of these, an LLM judge, nearly quadrupled false alarms, so it was kept off although it met its pre-fixed rule. Limitations: the labelled set is small (456 + 40 claims) and contains few Hindi hallucinations; live lookups need internet access; WHO data is licensed for non-commercial use only.

VII. Conclusion and Future Work

Satya-Score shows that a bilingual fact checker can be precise and transparent when it routes claims to authoritative sources, and that testing against rules fixed in advance exposes which seemingly attractive components do not help. Future work: a larger multi-domain Hindi evaluation set labelled on the current extractor, self-hosted knowledge sources, and LLM judgement restricted to a role that cannot add false alarms.

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